Numbers to 10: Ten-Frame Structure

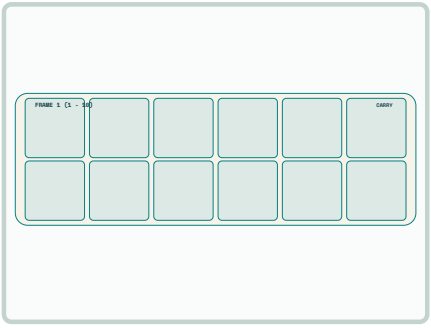
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.CC.B.5

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Fill top row (5 teal) then add coral counters in bottom row.

Sketch Model on Paper:
Draw filled and hollow circles in the 10-frame box.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	5	X: 5	Key 1st operand
2.	ENTER	Y: 5 | X: 5	Stack lift copies X
3.	2	Y: 5 | X: 2	Key 2nd operand
4.	+	X: 7	Drop stack & compute

EXPECTED LCD X: 7 counters

1 **Top row has 5, bottom row has 1. Total?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Top row has 5, bottom row has 3. Total?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Top row has 5, bottom row has 4. Total?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **A full ten-frame has 10 counters. Remove 3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 2

Numbers to 10: Ten-Frame Structure • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio (2x5 Grid) + Number Rack

CAD Blueprint: ten_frame_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/ten_frame_tray.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Frame with Coral & Teal Dots (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Top row has 5, bottom row has 1. Total?** Standard: CCSS.MATH.K.CC.B.5	6	[5] [ENTER] [1] [+]
2	**Top row has 5, bottom row has 3. Total?** Standard: CCSS.MATH.K.CC.B.5	8	[5] [ENTER] [3] [+]
3	**Top row has 5, bottom row has 4. Total?** Standard: CCSS.MATH.K.CC.B.5	9	[5] [ENTER] [4] [+]
4	**A full ten-frame has 10 counters. Remove 3.** Standard: CCSS.MATH.K.CC.B.5	7	[1][0] [ENTER] [3] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Ensure students recognize the full top row as 5 without counting individual pegs.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.